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Data Interpretation Exercise

Data Interpretation Assessment (R Programming)

Topics:

- Mapping Data onto Aesthetics
- Coordinate Systems and Axes
- Color Scales

Instructions: Use the given dataset to write an appropriate R program, generate the required visualization, and answer the questions.

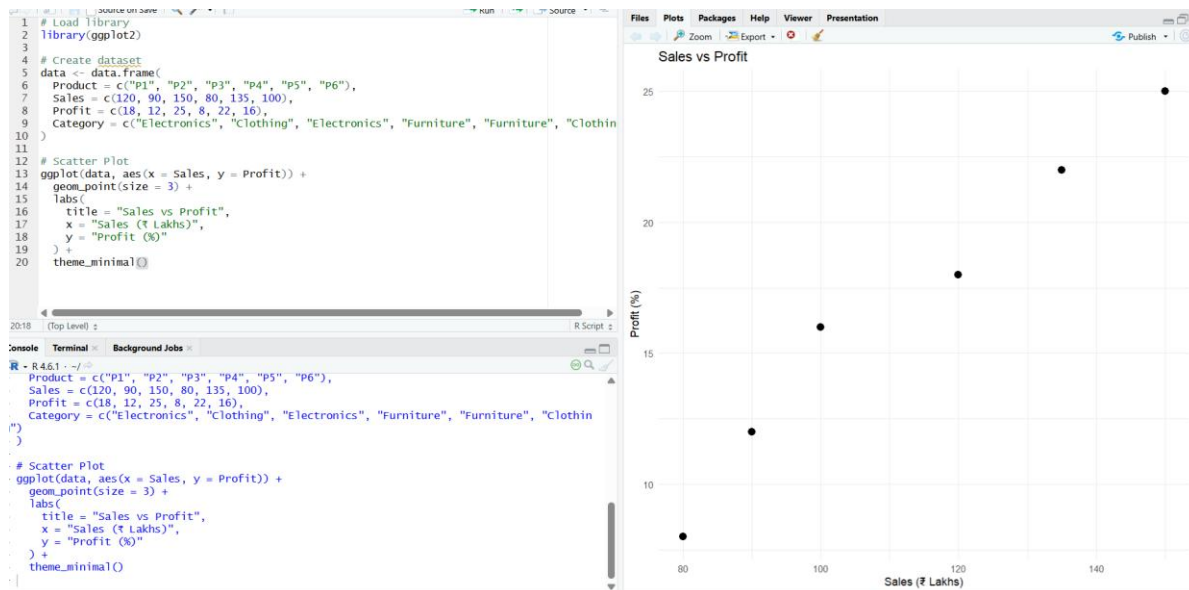
Question 1: Mapping Data onto Aesthetics

Dataset

Product	Sales (₹ Lakhs)	Profit (%)	Category
P1	120	18	Electronics
P2	90	12	Clothing
P3	150	25	Electronics
P4	80	8	Furniture
P5	135	22	Furniture
P6	100	16	Clothing

Questions

1. Write an R program to create a scatter plot with **Sales** on the x-axis and **Profit** on the y-axis.



2. Map **Category** using an appropriate aesthetic and **Sales** using another suitable aesthetic.

Answer:

- **Category** - Color (Different colors represent different categories such as Electronics, Clothing, and Furniture.)
- **Sales** - Size (Larger points represent higher sales, while smaller points represent lower sales.)

3. Interpret the relationship between Sales and Profit.

Answer:

There is a positive relationship between Sales and Profit. As Sales increase, Profit also tends to increase. Products with higher sales generally have higher profit percentages, indicating a positive correlation between the two variables.

4. Identify the type of scale used for each variable.

Answer:

Variable	Type of Scale
Sales	Continuous Scale
Profit	Continuous Scale
Category	Discrete (Categorical) Scale

5. Suggest one alternative visualization that better represents the same data.

Answer :

A **bubble chart** is an alternative visualization, where:

- **Sales** is shown on the x-axis,
- **Profit** on the y-axis,
- **Bubble size** represents Sales or Profit,
- **Color** represents the Category.

This helps compare multiple variable

Question 2: Coordinate Systems (Cartesian and Polar)

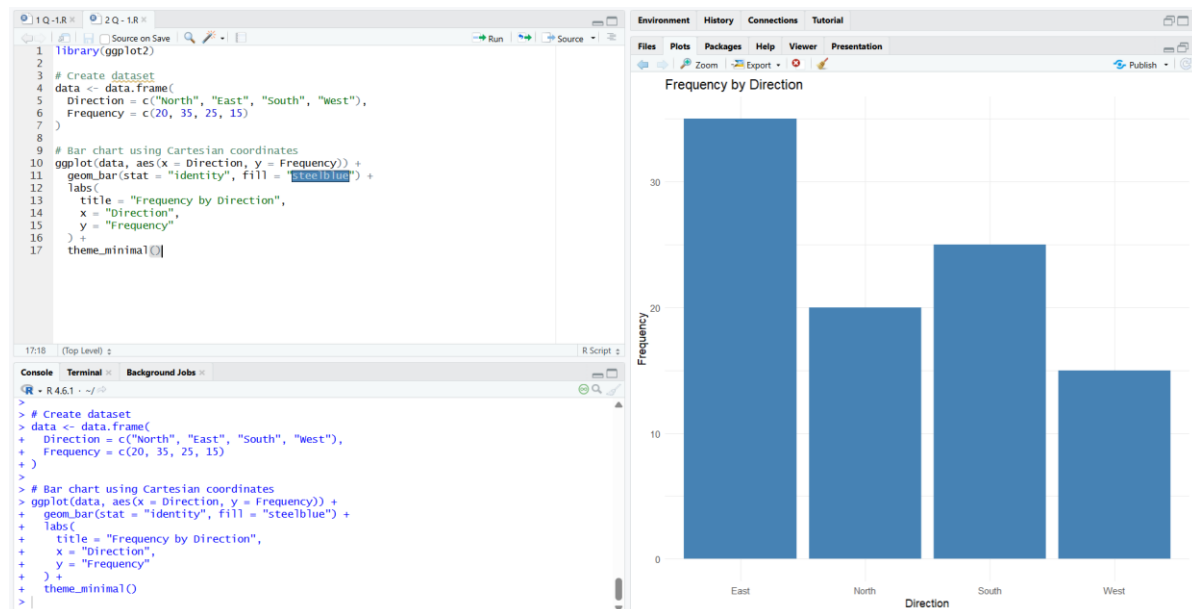
Dataset

Direction	Frequency
N	18
NE	12
E	20
SE	15
S	22
SW	11
W	16
NW	14

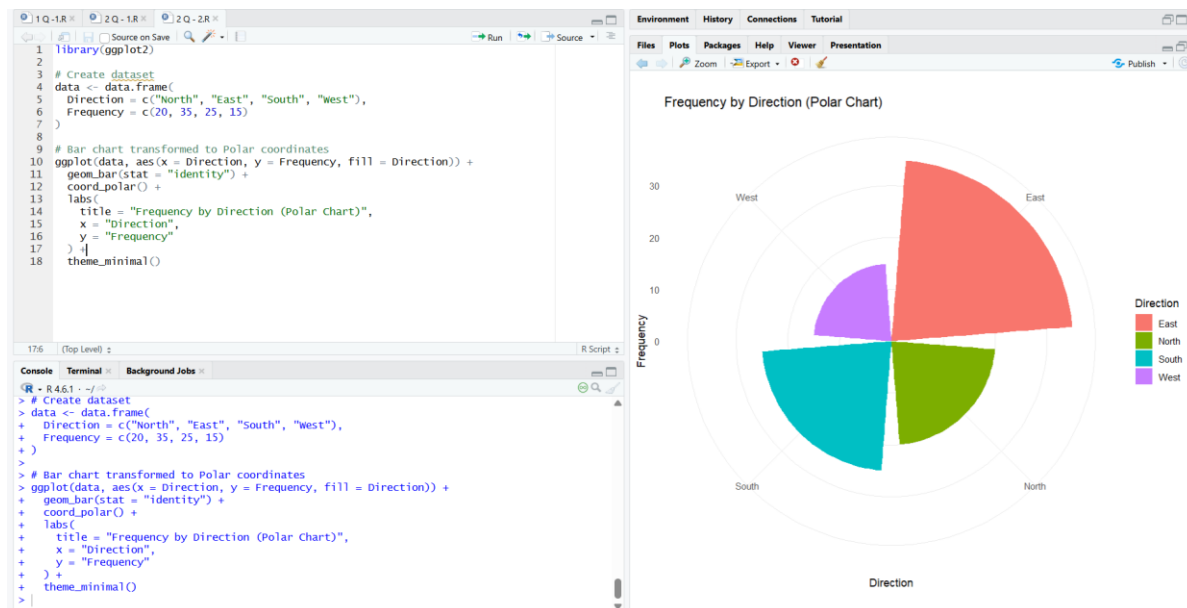
Questions

1. Write an R program to visualize the dataset using a Cartesian coordinate system.

Answer :



2. Write another R program to visualize the same data using a Polar coordinate system.



3. Compare the effectiveness of both visualizations.

Answer :

- **Cartesian coordinates:** The data is displayed as a bar chart with rectangular bars, making it easy to compare the frequencies.
- **Polar coordinates:** The data is displayed as a circular chart, emphasizing the distribution of frequencies around a circle and making it visually appealing for directional data.

4. Explain which coordinate system is more appropriate for directional data.

Answer:

The Polar coordinate system is more appropriate for directional data because it represents directions (North, East, South, West, etc.) around a circle, making the directional patterns easier to understand and interpret.

5. Justify your answer based on data interpretation.

Answer :

The Polar coordinate system is more appropriate because the data represents directions (N, NE, E, SE, S, SW, W, NW). In the polar chart, these directions are arranged naturally around a circle, making it easier to compare their frequencies. For example, South (22) has the highest frequency, while South-West (11) has the lowest, and this pattern is more intuitive to observe in a polar representation than in a Cartesian bar chart. Therefore, the polar coordinate system provides better interpretation for directional data.

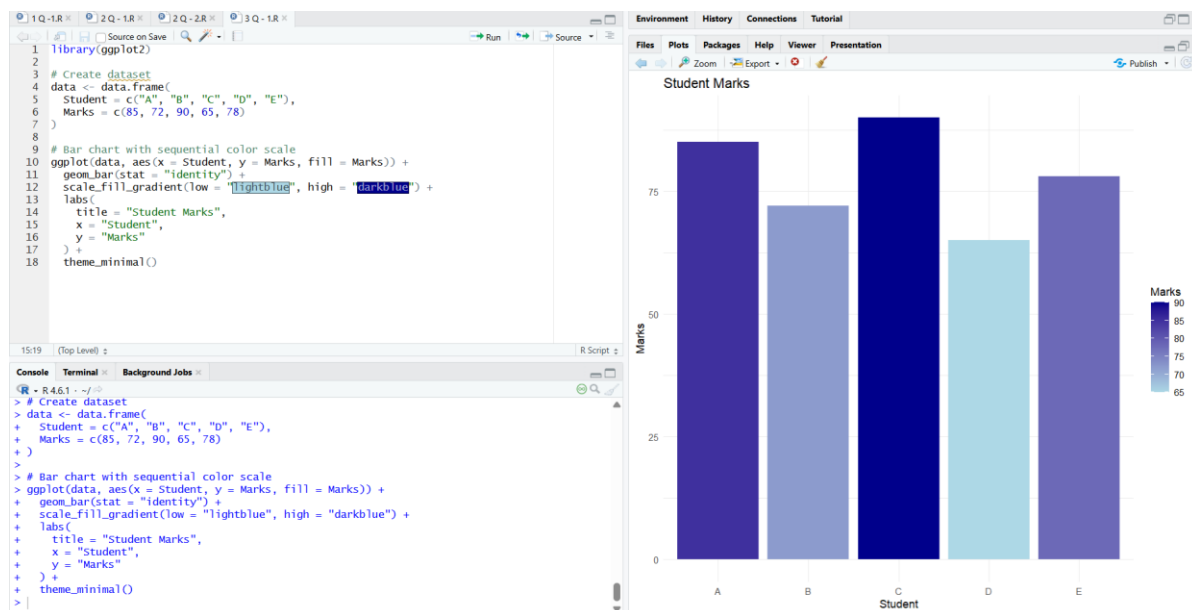
Question 3: Color Scales for Data Representation

Dataset

Student	Marks
A	45
B	58
C	72
D	81
E	67
F	90

Questions

1. Write an R program to create a bar chart using a sequential color scale based on Marks.



2. Identify the students with the highest and lowest marks from the visualization.

Answer :

Highest Marks: Student F – 90

Lowest Marks: Student A – 45

3. Explain why a sequential color scale is appropriate.

Answer :

A sequential color scale is appropriate because Marks is a continuous numerical variable. It uses lighter shades for lower marks and darker shades for higher marks, making it easy to compare students' performance and identify value differences.

4. Suggest a color palette suitable for color-blind users.

Answer :

A Viridis color palette is suitable for color-blind users because it provides clear color differences and is easily distinguishable for most types of color vision deficiency.

5. Explain the difference between sequential and categorical color scales.

Answer :

- **Sequential color scale:** Uses shades of a single color (light to dark) to represent ordered or continuous data, such as marks, sales, or temperature.
- **Categorical color scale:** Uses different colors to represent distinct categories with no inherent order, such as gender, department, or product category.

Question 4: Highlighting with Color

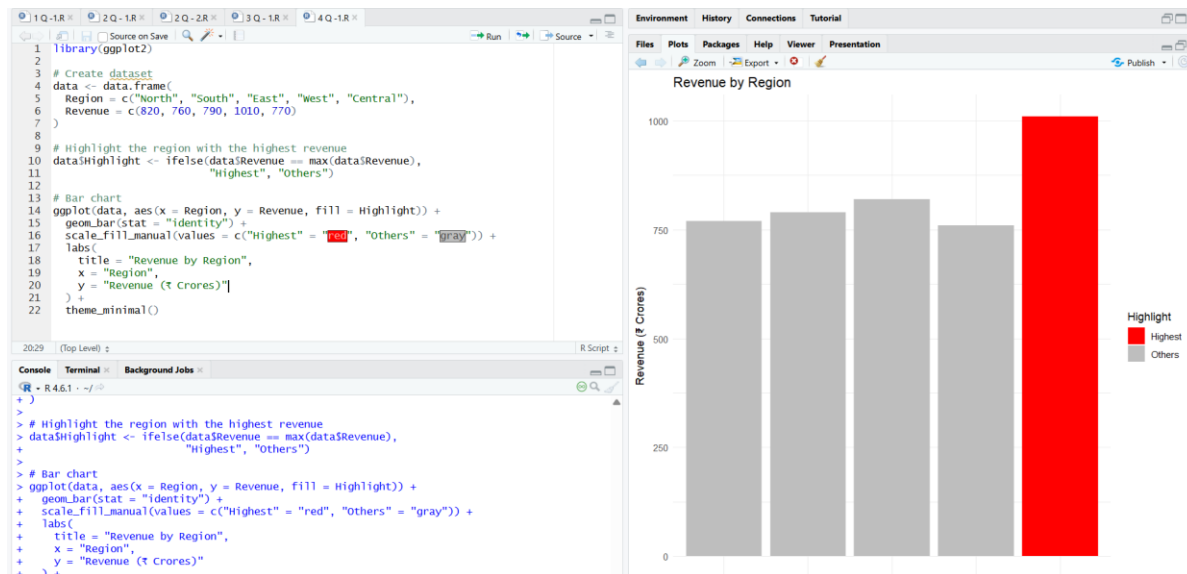
Dataset

Region	Revenue (₹ Crores)
North	820
South	760
East	790
West	1010
Central	770

Questions

1. Write an R program to create a bar chart highlighting the region with the highest revenue.

Answer :



2. Explain how highlighting improves data interpretation.

Answer :

Highlighting draws attention to important data points, making them easy to identify and compare. It helps users quickly focus on key information, such as the highest-performing region, without examining every value individually.

3. Identify the highest-performing region.

Answer :

The West region is the highest-performing region, with a revenue of ₹1010 crores.

4. Discuss how excessive use of colors can affect interpretation.

Answer :

Excessive use of colors can make the visualization cluttered and confusing. It may distract the viewer from the important information, make comparisons difficult, and reduce the effectiveness of highlighting key data points.

5. Suggest one alternative highlighting technique other than color.

Answer :

Use data labels to highlight the important value. For example, displaying the revenue value (₹1010 crores) above the West bar makes it easy to identify the highest-performing region without relying on color.

Question 5: Integrated Visualization

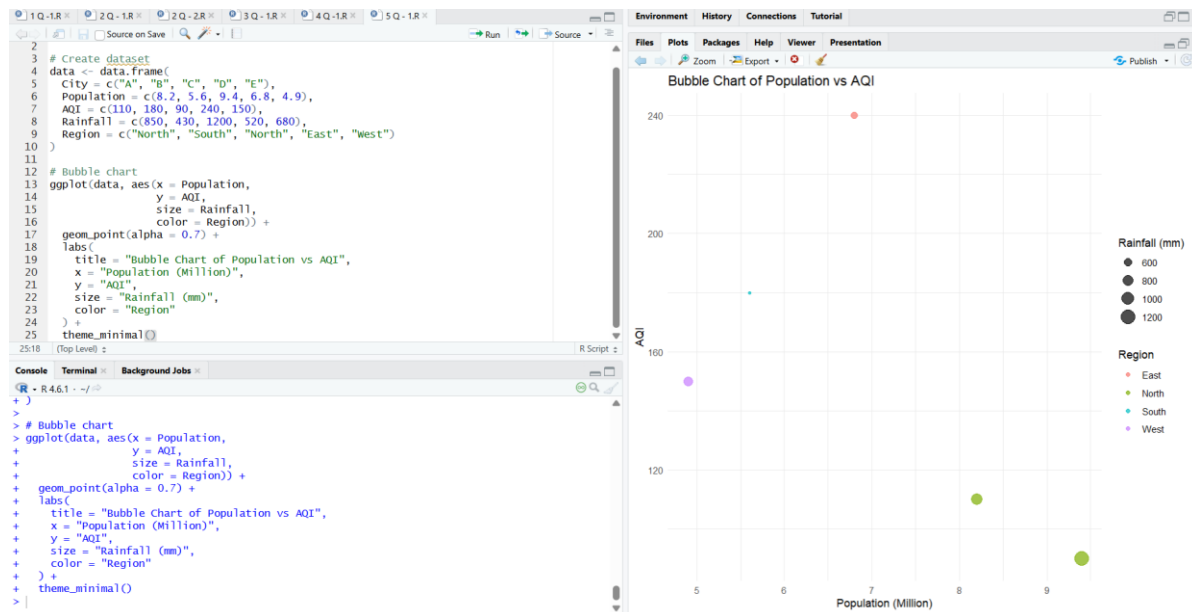
Dataset

City	Population (Million)	AQI	Rainfall (mm)	Region
A	8.2	110	850	North
B	5.6	180	430	South
C	9.4	90	1200	North
D	6.8	240	520	East
E	4.9	150	680	West

Questions

1. Write an R program to create a bubble chart using appropriate aesthetic mappings.

Answer :



2. Select suitable aesthetics for Population, AQI, Rainfall, and Region.

Answer :

- Population → X-axis
- AQI → Y-axis
- Rainfall → Bubble Size
- Region → Color

3. Interpret the relationship between Population and AQI.

Answer :

There is no clear linear relationship between Population and AQI. Some cities with higher population have lower AQI (e.g., City C), while others with lower population have higher AQI (e.g., City D). This indicates that factors other than population also influence AQI.

4. Identify the continuous and categorical variables.

Answer :

Continuous variables:

- Population (Million)
- AQI
- Rainfall (mm)

Categorical variables:

- City
- Region

5. Recommend one improvement to enhance the visualization's readability.

Answer :

Add labels for each city (A, B, C, D, and E) next to the bubbles so that each data point can be identified easily without referring back to the dataset.